

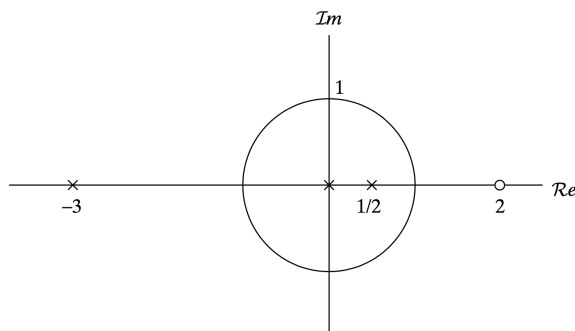
Homework Set #3

1. (3 pts) Consider a causal system

$$H(z) = \frac{5 - 5z^{-1}}{1 + 1.5z^{-1} - z^{-2}}.$$

Determine its:

- (a) impulse response, i.e., output $y[n]$ when the input is $x[n] = \delta[n]$;
 - (b) step response, i.e., output $y[n]$ when the input is $x[n] = u[n]$;
 - (c) difference equation relating input and output.
2. (4 pts) A bounded-input bounded-output stable discrete-time LTI system has impulse response $h[n]$ corresponding to a rational system function $H(z)$ with the pole-zero diagram shown below. [Note: $\times$ indicates poles, $\circ$ indicates a zero.]



Moreover, we know that $H(e^{j\pi}) = -\frac{1}{2}$.

- (a) Determine $H(z)$ and its ROC.
- (b) A new system is designed such that its impulse response is $f[n] = h[n] * h[-n]$. Find $F(z)$ and its ROC.

[Hint: Given a pole-zero diagram with pole p_0 and zero z_0 , the corresponding transfer function is $K \frac{z - z_0}{z - p_0}$, where K is a scaling coefficient that can be chosen to enforce a condition on $H(e^{j\omega})$.]

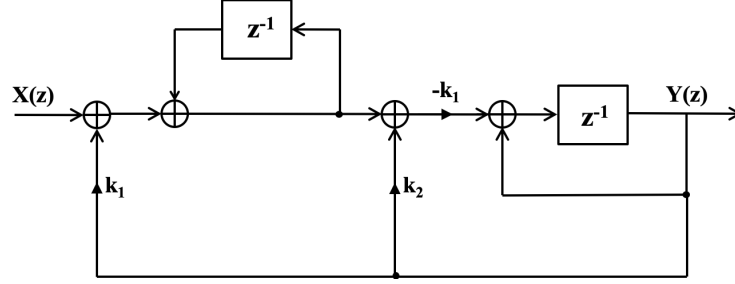
3. (3 pts) A stable discrete-time LTI system is described by the following difference equation:

$$y[n] - y[n - 1] + Cy[n - 2] = x[n],$$

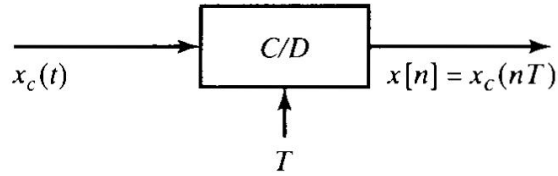
where C is a real number. Determine the range of C so that

- (a) the system is causal;
- (b) the system is anti-causal;
- (c) the system is non-causal (i.e., it has a two-sided impulse response).

4. (4 pts) A discrete-time LTI system is described by the following block diagram:

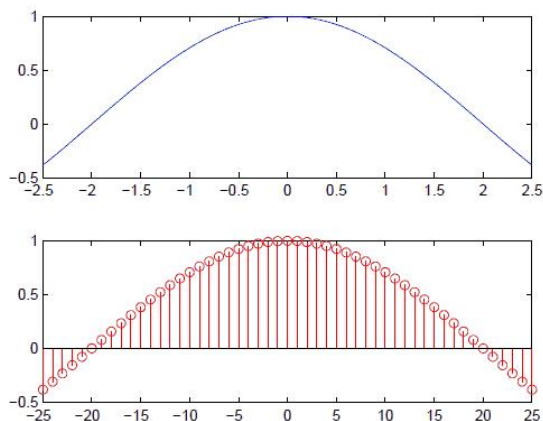


- (a) Find its transfer function $H(z)$. Specify the corresponding ROC for $k_1 = -\frac{1}{2}$, $k_2 = 2$.
- (b) Find the impulse response of the system $h[n]$.
5. (3 pts) Each of the following continuous-time signals is used as the input $x_c(t)$ to an ideal C/D converter as shown in the figure below with the sampling period T as specified. In each case, find the resulting discrete-time signal $x[n]$. Use Python to illustrate the original signal and its sampled version. (For details, please see the explanation on the next page).



- (a) $x_c(t) = \cos(2\pi(1000)t)$, $T = (1/3000)\text{s}$;
- (b) $x_c(t) = \sin(2\pi(1000)t)$, $T = (1/1500)\text{s}$;
- (c) $x_c(t) = \sin(2\pi(1000)t)/(\pi t)$, $T = (1/5000)\text{s}$.

Instructions for Problem 5: Say that we are given $x(t) = \cos(\pi/4t)$, and $x[n] = \cos(\pi/40n)$. You can verify that $x[n]$ can be obtained by sampling $x(t)$ with $T = 0.1$ s. Let us plot $x[n]$ for $-25 \leq n \leq 25$. For a comparison, we will also plot $x(t)$ for $-2.5 \leq t \leq 2.5$ (remember, we sample at $t = nT$). This is what it looks like:



For each of the parts in Problem 5, you should submit a graph such as the one above. Note: Be careful about sinc function (part c); to resolve $\frac{0}{0}$ issue, you need to declare $x[0]$ separately from the rest of the sequence $x[n]$.